

OS LAB EXPERIMENTS 2

PROGRAM 1:

```
nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ cat program1.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    printf("Before execl()\n");
    execl("/bin/ls", "ls", "-l", NULL);
    printf("This will not print if execl() succeeds.\n");
    return 0;
}

nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ ./program1
Before execl()
total 116
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:16 progarm1
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:17 program1
-rw-r--r-- 1 nishanth nishanth   187 Jul 31 09:15 program1.c
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:18 program2
-rw-r--r-- 1 nishanth nishanth   182 Jul 31 09:18 program2.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:21 program3
-rw-r--r-- 1 nishanth nishanth   210 Jul 31 09:20 program3.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:22 program4
-rw-r--r-- 1 nishanth nishanth   203 Jul 31 09:22 program4.c
-rwxr-xr-x 1 nishanth nishanth 16120 Jul 31 09:24 program5
-rw-r--r-- 1 nishanth nishanth   243 Jul 31 09:23 program5.c
```

PROGRAM 2:

```
nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ cat program2.c
#include <stdio.h>
#include <unistd.h>
int main()
{
    printf("Before execlp()\n");
    execlp("date", "date", NULL);
    printf("This will not print if execlp() succeeds.\n");
    return 0;
}

nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ ./program2
Before execlp()
Fri Jul 31 09:29:46 UTC 2026
```

PROGRAM 3:

```
nishanth@LAPTOP-6Q30N07H:~/OSLAB/exec$ cat program3.c
#include <stdio.h>
#include <unistd.h>
int main()
{
char *args[] = {"ls"
,
"-l"
, NULL};
printf("Before execv()\n");
execv("/bin/ls"
, args);
printf("This will not print if execv() succeeds.\n");
return 0;
}
```

```
nishanth@LAPTOP-6Q30N07H:~/OSLAB/exec$ ./program3
Before execv()
total 116
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:16 program1
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:17 program1
-rw-r--r-- 1 nishanth nishanth  187 Jul 31 09:15 program1.c
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:18 program2
-rw-r--r-- 1 nishanth nishanth  182 Jul 31 09:18 program2.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:21 program3
-rw-r--r-- 1 nishanth nishanth  210 Jul 31 09:20 program3.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:22 program4
-rw-r--r-- 1 nishanth nishanth  203 Jul 31 09:22 program4.c
-rwxr-xr-x 1 nishanth nishanth 16120 Jul 31 09:24 program5
-rw-r--r-- 1 nishanth nishanth  243 Jul 31 09:23 program5.c
```

PROGRAM 4:

```

nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ ./program3
Before execv()
total 116
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:16 progarm1
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:17 program1
-rw-r--r-- 1 nishanth nishanth  187 Jul 31 09:15 program1.c
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:18 program2
-rw-r--r-- 1 nishanth nishanth  182 Jul 31 09:18 program2.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:21 program3
-rw-r--r-- 1 nishanth nishanth  210 Jul 31 09:20 program3.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:22 program4
-rw-r--r-- 1 nishanth nishanth  203 Jul 31 09:22 program4.c
-rwxr-xr-x 1 nishanth nishanth 16120 Jul 31 09:24 program5
-rw-r--r-- 1 nishanth nishanth  243 Jul 31 09:23 program5.c
nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ cat program4.c
#include <stdio.h>
#include <unistd.h>
int main()
{
char *args[] = {"date"
, NULL};
printf("Before execvp()\n");
execvp("date"
, args);
printf("This will not print if execvp() succeeds.\n");
return 0;
}
nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ ./program4
Before execvp()
Fri Jul 31 09:31:12 UTC 2026

```

PROGRAM 5:

```
nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ cat program5.c
#include <stdio.h>
#include <unistd.h>
extern char **environ;
int main()
{
char *args[] = {"ls"
,
"-l"
, NULL};
printf("Before execve()\n");
execve("/bin/ls"
, args, environ);
printf("This will not print if execve() succeeds.\n");
return 0;
}
nishanth@LAPTOP-6Q3ON07H:~/OSLAB/exec$ ./program5
Before execve()
total 116
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:16 program1
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:17 program1
-rw-r--r-- 1 nishanth nishanth 187 Jul 31 09:15 program1.c
-rwxr-xr-x 1 nishanth nishanth 16000 Jul 31 09:18 program2
-rw-r--r-- 1 nishanth nishanth 182 Jul 31 09:18 program2.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:21 program3
-rw-r--r-- 1 nishanth nishanth 210 Jul 31 09:20 program3.c
-rwxr-xr-x 1 nishanth nishanth 16048 Jul 31 09:22 program4
-rw-r--r-- 1 nishanth nishanth 203 Jul 31 09:22 program4.c
-rwxr-xr-x 1 nishanth nishanth 16120 Jul 31 09:24 program5
-rw-r--r-- 1 nishanth nishanth 243 Jul 31 09:23 program5.c
```